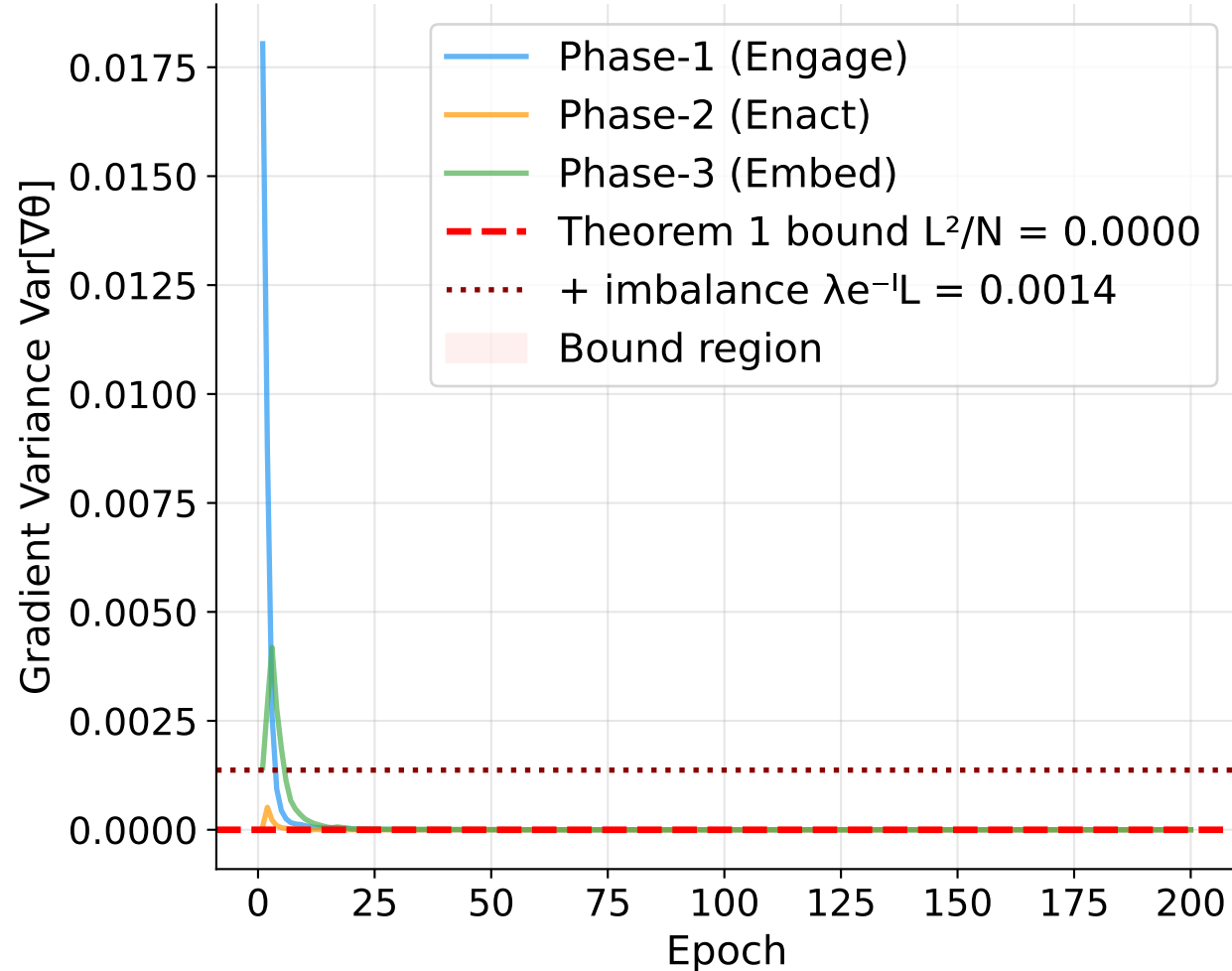
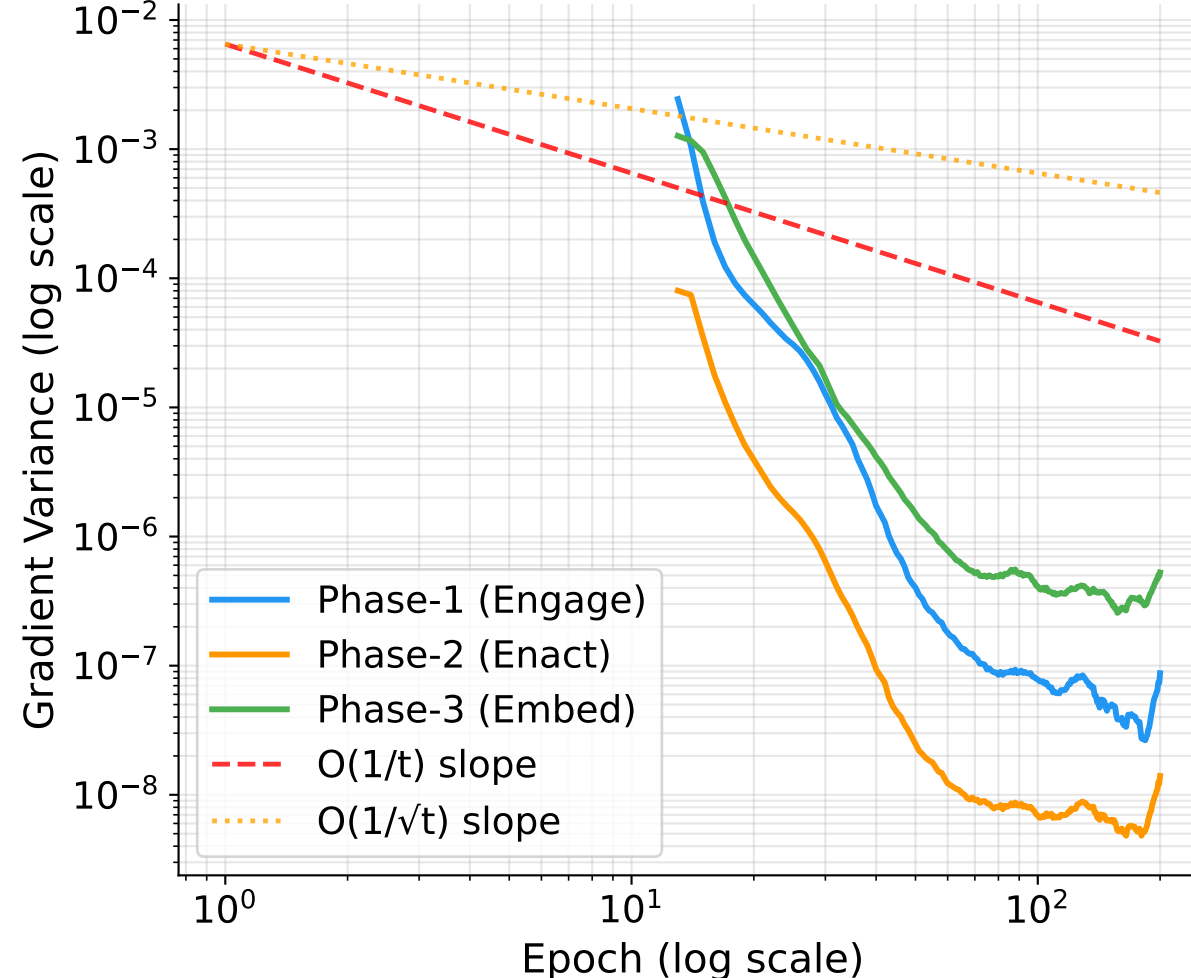


Empirical Variance vs Theorem 1 Bound
 $\text{Var}[\nabla] \leq L^2/N$



Variance Decay Rate — Log-Log
Confirms $O(1/t)$ convergence rate (Theorem 1)



Imbalance Penalty Validation
Engage phase excess $\leq \lambda e^{-L}$ (Theorem 1 term 3)

